



PFAS-Free Laser Textured Patterned Wetting Surface for Solar-Powered Fog Harvesting



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Motivation & Fundamental Theory

Motivation

- ❑ Fresh water scarcity affects billions of people
- ❑ Fog harvesting offers a passive, low-energy water collection method [1]
- ❑ Many high-performance surfaces rely on PFAS-based coatings
- ❑ This work develops a PFAS-free, laser-treated, wettability-patterned aluminum surface with solar-powered cooling to improve water collection

Fundamental Theory

- ❑ Fog harvesting involves three key steps: capture, collect, and transport of condensed droplets
- ❑ Uniform superhydrophilic surfaces promote droplet spreading but hinder transport due to strong adhesion, while uniform superhydrophobic surfaces enable removal but limit controlled collection
- ❑ Wettability-patterned surfaces integrate surface energy gradients, generating a driving force that directs droplets toward more wettable regions
- ❑ Geometric features [2] (e.g., tips or wedge-like structures) create Laplace pressure gradients, further accelerating directional droplet motion
- ❑ Integrating wettability contrast and shape-induced pressure gradients improves overall fog collection efficiency beyond uniform surfaces

Fundamental of water capture, collection and transport

- ❑ Wettability gradient driving force [3]

$$F_{wet} \approx \gamma_{LV} (\cos \theta_{hydrophilic} - \cos \theta_{hydrophobic})$$

Where,

$\theta_{hydrophilic}$ = contact angle on hydrophilic region

$\theta_{hydrophobic}$ = contact angle on hydrophobic region

γ_{LV} = water surface tension

- ❑ Laplace pressure (curvature effect) [3]

$$\Delta P = \gamma_{LV} \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

Where,

ΔP = internal pressure difference

R_1, R_2 = principal radii of curvature

γ_{LV} = water surface tension

Experimental Methodology

Surface Samples

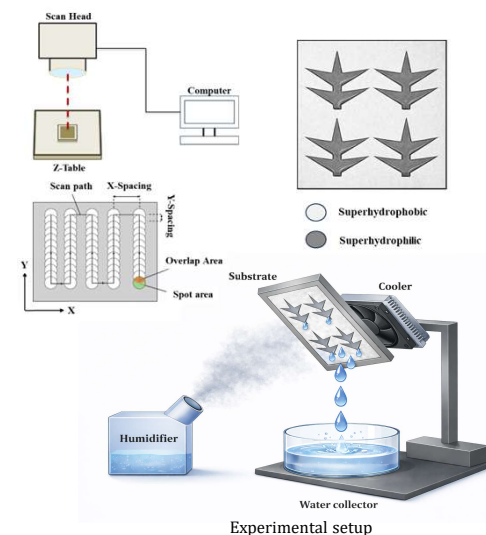
- ❑ Three aluminum samples: untreated, superhydrophobic, and patterned
- ❑ The superhydrophobic surface was fabricated by laser texturing followed by PFAS-free chemical treatment
- ❑ The patterned surface was created by selective re-laser of the superhydrophobic surface and subsequent boiling water treatment to grow hydrophilic boehmite regions

Experiment Setup

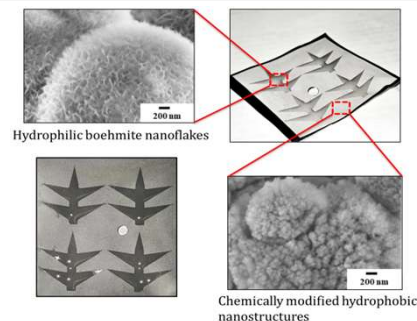
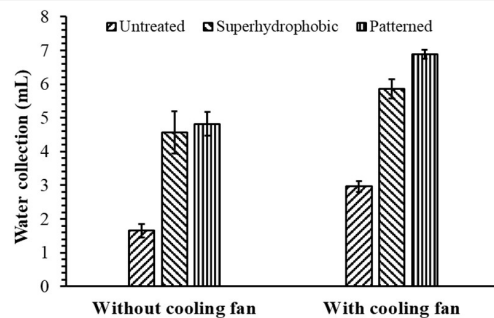
- ❑ Commercial humidifier set at 45° angle was used to generate fog
- ❑ Samples and Peltier module were mounted at 45°, ensuring perpendicular fog impact
- ❑ Distance between humidifier outlet and sample surface: 70 mm
- ❑ Fog flow rate and humidifier settings were kept constant throughout all experiments

Experimental Conditions

- ❑ A Peltier cooler was mounted behind the aluminum surface
- ❑ Cooling intensity was controlled using two conditions:
 - ❑ No cooling
 - ❑ Peltier cooler active
- ❑ Both treated and untreated surfaces were tested under these two conditions
- ❑ Collection time and all other parameters were kept constant



Results and Discussions



- ❑ Activating Peltier cooling increased water collection on the untreated aluminum surface from 1.657 mL to 2.964 mL, corresponding to an increase of approximately 79%
- ❑ For the fully superhydrophobic surface, cooling increased water collection from 4.566 mL to 5.861 mL, an improvement of approximately 28%
- ❑ The patterned superhydrophilic-superhydrophobic surface increased from 4.818 mL to 6.884 mL, corresponding to an improvement of approximately 43% with cooling
- ❑ The patterned surface achieved the highest overall water collection under both cooling and no-cooling conditions

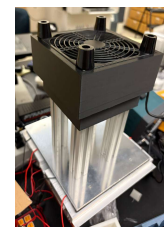
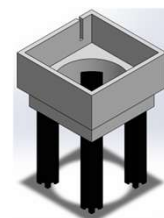
Conclusions and Future Work

Conclusion

- ❑ Surface engineering significantly improved fog harvesting performance. Without cooling, the superhydrophobic (4.566 mL) and patterned (4.818 mL) surfaces collected nearly three times more water than untreated aluminum (1.657 mL)
- ❑ Active cooling further enhanced performance across all surfaces; under cooling, the patterned surface achieved the highest collection (6.884 mL), demonstrating that integrating wettability patterning with thermal control provides the greatest fog harvesting efficiency
- ❑ Integrating wettability contrast with active cooling provides a clear advantage for fog harvesting

Future work

- ❑ Integrate the treated surface and cooling system into a portable water harvesting device
- ❑ Optimize pattern geometry and hydrophilic area fraction
- ❑ Cool all internal surfaces using Peltier coolers to enhance condensation
- ❑ Develop a fully solar-powered, off-grid fog harvesting system
- ❑ Evaluate long-term durability and outdoor performance



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[2] Samanta, A., Huang, W., Ding, H. *Efficient fog harvesting on metal surfaces with venation-inspired wetting patterns*. J. Manuf. Process., 2023.
[3] Bai, H., Wang, L., Ju, J., Sun, R., Zheng, Y., Jiang, L. *Efficient Water Collection on Integrative Bioinspired Surfaces with Star-Shaped Wettability Patterns*. Adv. Mater., 2014.

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